Journal of King Saud University – Computer and Information Sciences 33 (2021) 1–10

Contents lists available at ScienceDirect

Journal of King Saud University – Computer and Information Sciences

journal homepage: www.sciencedirect.com

Perusal of readability with focus on web content understandability

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ARTICLE INFO

Article history:
Received 25 October 2017
Revised 29 January 2018
Accepted 7 March 2018
Available online 21 March 2018

Keywords:
Easy to Read (E2R)
Readability
Understandability
Readability models
Readability indices

ABSTRACT

The Web has become a popular and important medium of transmitting information from one place to another place. For making information accessible to all, we need to check their accessibility and readability score. Readability is a metric to measure the success of information being successfully conveyed to a large population when people are trying to access it. This survey paper analyzes the popular readability indices used to check the readability of websites and web pages. Most of the available readability metrics consider only textual features of the web while checking the readability of websites, and a lot of factors which affect readability are not taken into account. Based on survey findings, we provide some useful suggestions, which are, if considered while developing readability formulas, the results will be more effective.

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1. Introduction

Nowadays services and information are accessible to the users with huge nodes of a network called *World Wide Web* (WWW). People from different cultural and linguistic backgrounds meet through this global environment for sharing information and services without any geographical barriers. If a user is able to get the information with out any barrier then the interface and contents are treated as understandable. Language is a medium of communication. It is highly desirable that the language should be as much straightforward, plain and simple. The usage of language with aforementioned characteristics make the design of the interface quick to understand and navigate easily. These properties makes such site inclusive and bring a large group of users to the resource. The collection of all elements in textual material that affects level of understanding of the reader constitutes the *readability* components. Readability is a function of reader's educational and social background along with his/her expertise and motivation to learn.

The *WCAG 2.0 principles and Easy-to-Read (E2R) guidelines* provide guidelines and suggestions that needs to be considered while creating the web contents for inclusive design of websites. Technical aspects of web content includes understandability, readability, memorability that need to be considered to target

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Peer review under responsibility of King Saud University.

Production and hosting by Elsevier

https://doi.org/10.1016/j.jksuci.2018.03.007
1319-1578/

people with disabilities and make more web contents user friendly for them.

A web content can be considered as accessible if it follows the WCAG 2.0 guidelines. Then, use of simple language makes an information usable, readable and understandable for the readers. Thus, Web contents based on plain language, WCAG 2.0 and Easy-to-Read guidelines, targets a large group of people with high understandability.

There are various assessment methods used so far to assess the readability of text and web contents. Using these assessment methods for evaluation processes have been an active area of research because low readability directly affect understandability of the web. Some brief literature work based on readability tools are presented in Subsection 1.1.

The primary focus of this paper is to present an analysis of various readability computation techniques. Along with this we have considered the criticality of readability in web resources, especially when it is accessed by persons with disabilities. In this study, we have framed four objectives for analyzing, comparing and identifying the attributes of readability indexes, as presented in Subsection 1.2.

1.1. Readability research studies

Readability analysis is an active area of research with contributions from various dimensions (Miltsakaki, 2007; Broda et al., 2014; Begeny and Greene, 2014; Biddinika et al., 2016). Information that is available on web pages should be able to reach wider section of users, easy-to-read and easy-to-understand are the aspects essential for making the pages universally accessible. These aspects needs to be considered along with aspects of technical accessibility (Matausch et al., 2014). The major objective of "Easy-to-Read" on the web is to collect concise and up-to-date recommendations and also to raise awareness of problems which the users especially cognitive disabilities are experiencing.

Readability is an issue which has been discussed long in measuring text complexity. Despite a conclusive and fully representative scheme with respect to measure the readability, which can provide computational criteria to evaluate the complexity of the text is still not presented (Yasseri et al., 2012). A readability assessment method which hypothesized that readability is determined on the basis that a reader can easily understand text structures, a method was presented by researchers (Yamasaki and Tokiwa, 2014) to assess readability rank using machine learning methods.

A Latent Trait Model is applied to a subset of variables, gathered in a reading experiment to predict reading difficulty by the use of automatic readability assessment tool for people with mild intellectual disabilities (Jansche et al., 2010). For people with cognitive impairments, a text need to be simplified as text may be complex and difficult to read for cognitively disabled people.

Researchers (Lasecki et al., 2015) proposed to measure the simplicity of the text by the use of crowd. Use of 2500 crowd annotations shows that crowd can be effectively used to rate levels of simplicity of text (Lasecki et al., 2015). Another research targets to develop a web-based automated system that can be used to simplify text for people with intellectual disabilitiesby considering the challenges faced in understanding written and spoken information (Huenerfauth et al., 2009). A study was carried out by researchers (Chung et al., 2013) on Deaf people. Deaf people faced a lot of difficulties to understand text-based web documents because of visual orientation of their sign language. A system was developed (Chung et al., 2013) that converts complex sentences to simple sentences and present the relations among them (Deaf people)with graphical representation toenhance the readability of web documents.

A research work was carried out by researchers like (Ismail et al., 2019; Ismail and Kuppusamy, 2016; Ojha et al., 2018) on accessibility, readability and site ranking of websites by using different accessibility and readability evaluation tools (Ismail et al., 2017). They also find their correlations in order to connect the accessibility, readability and ranking of sites to each other. Another work was carried out on readability and accessibility of Indian universities homepage's by using different evaluation tools like achecker and wave tool for accessibility, and Gunning fog for readability (Ismail and Kuppusamy, 2018).

As an experiment to test whether the existing English readability metrics are applicable for other languages or not was carried out for Bangla language (a popular language spoken in India). It was found inapplicable (Sinha and Basu, 2014). For this study, they (Sinha and Basu, 2014) have used different machine learning methods like regression, support vector machines (SVM) and support vector regression (SVR). Due to large technological advancement in the field of readability, Rebekah George Benjamin presented (Benjamin, 2012) a review focused on recommendations for current and future research because almost all the fields of education are developing methods for predicting text difficulty.

A metric-based tool named GUI Evaluator is presented for evaluation of complexity of user interfaces based on structural measures (Alemerien and Magel, 2014). Role of language in accessing information assessed and suggests that language may act as a double barrier. Web-hosts and links-data obtained using a crawler and details of website users through log-file analysis and results matched with *Information Foraging Theory* and *Revised Hierarchy Model* (Kralisch and Mandl, 2006). A set of protocols recommends and assist professionals to identify characteristics and problems while evaluating interfaces using totally impaired vision people, which aims to identifying the usability problems (Ferreira et al., 2012). The designers of the interfaces should be able to analyze the needs of users by their involvement during designing (Ferreira et al., 2012).

1.2. Research objectives

Language plays the most important role in communicating the information and hence enough emphasize need to be given in presenting the text with better readability.

The objectives of this paper are as listed below:

1. To analyze the existing readability methods which are used for language understand-ability.
2. To present the workflow and comparative analysis of the existing readability methods.
3. To present the existing readability approaches for web contents.
4. To suggest missing links in the computation of readability score for web contents by incorporating web specific features.

2. Analysis of traditional readability

During our survey, a list of readability models and tools, which we have studied and analyzed are presented in Table 1. The models developed by various researchers aimed to estimate the readability of different written materials. We analyzed that most of the formulas estimated the readability in terms of grading level based on US grading system. The primary and secondary grading level varies on geographic location and environment, a question is unanswered till yet on the validity of these formulas globally. Djoko formula, Fernandez Huerta Index, Kandal and Moles Index and Al-Heeti grade level are various models developed by researchers to predict readability of Indonesian text, Spanish text, French text and Arabic text respectively. The formula developed for foreign text(other than English) proved correlated with the existing reading capabil-

Table 1
Traditional readability methods.

S. No	Tool/Model	Basis of Readability Formula	Inferences
1	The Flesch Read-ing Ease Read-ability Formula (RudolphFlesch (1948))	206.83−(1.015∗ASL,) − (84.6 ∗ ASW,)	One of the accurate measure for school text. Higher the score, text is easily readable, difficulty increases with decrease in score level. Score ranges from 0 to 100.
2	Dale-Chall Read-ability Formula (Edgar Dale and JeanneChall (1948)) (Dale and Chall, 1948)	0.1579∗(PDW,) + (0.0496 ∗ ASL)	Inspired by FRE, considers difficult words and sentence length to result Dale-chall score. A score below 4.9 for a text is treated as easily understood by Grade 4 student in US and grade above 10 for college graduates.
3	Flesch- Kincaid Grade Level ReadabilityFormula (Kincaid et al., 1975) (Rudolf Flesch and John P. Kincaid (1976))	(0.39 ∗ ASL) + (11.8 ∗ ASW) − 15.59	Modified version of FRE, Score indicates the grade-school level education required to understand textual content in US.
4	FOG (Gunning, 1952) (Robert Gunning (1952))	0.4 ∗(ASL + PHW)	Research based on daily newspapers, magazines is the origin of this formula, text with score 7–8 Fog Index is considered as ideal and score above 12 is too hard for most of the people.
5	Forcast (Begeny and Greene, 2014) (John S Caylor, Thomas G Sticht, and J PatrickFord (1973))	GL = 20−(n/10) Age to Read = 25−(n/10) years	#n = Number of Single Syllable word. Considered as perfect formula for multiple choice questions related text material in US. Strictly not to be used to assess primary age reading materials.
6	Fry (Fry, 1990) (Edward Fry (1968))	Based on graph plotted mean value of sentences per 100 words versus mean value of syllables per 100 words.	Used to provide consensus of readability of regulatory purposes. Scores are probably within a grade level.
7	PSK (Powers-Sumner-Kear) ReadabilityFormula (Begeny and Greene, 2014) (R D Powers, W A Sumner, and B E Kearl (1958))	gl = 0.0778(ASL) + 0.0455(ns) − 2.2029 ra = 0.0778(ASL) + 0.0455(ns) + 2.7971	#ns = Number of syllables,gl = grade level ra = Reading Age. Best formula to calculate text sample for US grade level, suited best for primary age children and advised not to use for text for children above age of 10.
8	Automatic Readability Index (Smith and Senter, 1967)	4.71 ∗ ACW + 0.5 ∗ AWS−21.43	Outputs an approximate grade level required to comprehend the text, for example US grade Level 1 corresponds to understandable to age 6 to 8 and grade 12 corresponds to a 17 year old. It is based count of characters and words.
9	CLI(Meri Coleman and T.L. Liau)	0.0588L-0.296S-15.8	It approximates US grade level to understand the text based on characters instead of syllables. A grade level of 10.6 is easily understood by 10–11 grade student whereas 14 is for college level student.
10	BRI (John R. Bormuth)	0.8865 − (AWL ∗ 0.036) + (AFW ∗ 0.161911) − (ASL ∗ 0.21401) − (ASL ∗ 0.000577) − (ASL ∗ 0.000005)	To count familiar words BRI uses Dale-Chall word list (Dale and Chall, 1948) in samples of text. It closely matches new Dale-Chall readability formula, the only difference is, it relies on count of characters instead of count of syllables and considers average familiar words instead of percent difficult words.
11	LIX (Formula, 2017) (Carl-Hugo Bjornsson 1968)	$\frac{W}{Pr} + 100 * \frac{D}{W}$	It is the formula used to predict readability of texts in French language. A lix score of 20 to 25 is considered as very easy and a score of 60 is considered as very difficult.
12	Raygor Estimate Graph (Alton L. Raygor 1977)	Based on graph plotted for average number of sentences (Y-axis) versus average number of characters (count more than 6 on X-axis).	The intersection point of X-Y axis represents the grade level and grade level is valid if point of intersection is within the parallel lines otherwise invalid. Grade level ranges between 3 to 14.
13	Djoko Formula (Biddinika et al., 2016) (D. Djoko Pranowo)	JKT-4 = Amount of scores for all indicators.13 indicators based on paragraph, words and sentences	Djoko formula which is based on 13 indicators of a text (paragraph,words and sentences) is used to categorize readability of Indonesian text. Criterion range is done by looking difference between easy text and hard text.
14	Pisarek's Index (Broda et al., 2014)	$P1 = \frac{1}{3} * ASL * \frac{1}{3} * PCW + 1$ $P2 = \frac{1}{2} * \sqrt{ASL^2 + PCW^2}$	It is similar to FOG index based on average length of sentence and percentage of complex words. P1 and P2 are two different versions of this formula one is linear and other is non-linear.
15	The Mistrik Formula (Gavora, 2012) (Jozef Mistrik 1982)	$50 - \frac{As*Av}{I}$	Three parameters required to compute text readability. As = Average length of words in number of syllables Av = Semantic difficulty by average length of sentences I = variability of words
16	Fernandez Huerta Index (Fernández Huerta, 1959) (1959)	206.84 − (0.60 ∗ p) − (1.02 ∗ f) p = number of syllables f = number of sentences contained in 100 words	Still widely used formula to calculate readability of Spanish Text. It is an adaptation of FRE. Huerta formula is not scalable in its original form.
17	Kandal & Moles Index (François and Fairon, 2012) (1958)	207–1.015Lp − 0.736Lm	It is an adaptation of FRE for French text. Lp = Average number of words per sentence Lm = average number of syllables per word
18	Al-Heeti Grade Level (Al Tamimi et al., 2014)	AWL ∗ 4.414 − 13.468	Al-Heeti readability formula results a score indicating the grade level r to comprehend arabic text.
19	Test Evaluator(Napolitano, 2015) (Diane Napolitano et al.)	Tool based on sentence structure, vocabulary difficulty, connection across ideas and degree of narrativity.	A tool capable of analyzing any written text, providing detailed information of readability of text and complexity.

(continued on next page)

Table 1 (*continued*)

S. No	Tool/Model	Basis of Readability Formula	Inferences
20	SMOG (Begeny and Greene, 2014) (G Harry McLaughlin (1969))	$3 + \sqrt{PolysyllableCount}$	SMOG predicts two grades higher than Dale-Chall formula. It is considered appropriate for secondary age. A text with polysyllabic count 1–6 falls under grade level 5 whereas 211–240 results a grade of 18.
21	Spache (Begeny and Greene, 2014) (G. Spache (1953))	$(0.141 * \mathrm{ASL}) + (0.086 * \mathrm{PDW}) + 0.839$	Spache is similar to Dale-Chall readability formula but not ideal for advanced texts (above grade 4)
22	The LexileFramework (Smith et al., 1989)	Based on Semantic Units (WordFrequency) and Syntactic Structures (Sentence Length)	It is limited with continuous prose but correlated with complexities while reading a comprehension, a lexile score correctly predicts comprehension ability of a person.
23	Advantage-TASA openStandard forReadability (ATOS)	Based on average number of characters in a word, Average number of Words in a Complete Sentence and Mean GradeLevel of Words along withbook length	ATOS uses large data bank of student reading performance in its development, version available for non-fictional textual matter along with scale of conversion for reading recovery levels.
24	Read-X (Miltsakaki, 2007)	Based on Number of sentences,number of words and letters,number of long words in text.	Read-X performs readability analysis of text on web which is real time based, performing web search and filtering by category level and categorizing results by theme.

ASL ⇒ Average Sentence length, ASW ⇒ Average Syllables per Word, PDW ⇒ Percentage Difficult Words, PHW ⇒ Percentage Hard Words, ACW ⇒ Average Complex Words, L ⇒ Mean number of letters per 100 words, S ⇒ Mean number of sentences per 100 words, AFW ⇒ Average number of familiar words, AWL ⇒ Average length of word, W ⇒ Count of Words, Pr ⇒ count of period, capital alphabet first and colon, D ⇒ count of words more than 6 characters.

ities of people (Biddinika et al., 2016; Fernández Huerta, 1959; François and Fairon, 2012; Al Tamimi et al., 2014), whereas scores resulted by tools for the English language were not in accordance with expectation.

Most of the readability formulas developed earlier, considered the factors such as length and count of words, sentences, syllables and complex words, which has a limitation that they may result in a good readability score to a nonsense text also. Only after the 1980s, the development of tools such as Lexile Framework, ATOS, Read-X, Coh-metrix and new Dale-Chall readability formula came into existence, which assess readability including factors such as cognitive-structural elements, semantic units and syntactic structures complexity.

The Lexile Framework is popular method for text leveling, text developed during the 1980s is complex in design. The Lexile Framework (Stenner, 1996) for reading is a unique resource to check ability to respond to comprehension questions correctly. In this framework, along with the score for reader, lexile score is also generated for text. If the reader has an accurate matching score as of text, the reader has the ability to answer comprehension question correctly. It evaluates the ability of reading based on actual assessments instead of generalized age or grade levels. Determination of reading comprehension is based on the familiarity of semantic units and syntactic structures complexity used in the matter. It includes the measure of frequency of word as a semantic variable and length of sentence as a proxy for complexity of syntactic features.

Renaissance Learning Inc. and Touchstone Applied Science Associates (TASA) Inc. created two formulas using reading assessment database and massive book. The formulas named as *ATOS for books readability* and *ATOS for text readability*. Length of word and sentence along with grade level of words are the traditional variables over which both the formulas are based and one factor which influences difficulty of book is the length of the book that needs to be considered for the formula of books. Development of the formula for book-matching process considers the weaknesses of formulas which were developed earlier (Benjamin, 2012). The following areas were identified which were improved.

1. The basis for the semantic aspect of a readability formula is updated by broadening the corpus of words.
2. Different kinds of texts need to be expanded for which the process would be appropriate.
3. Possible adjustments need to be considered which accounts for repetition of special words within the text.

To guide students appropriate level books, the ATOS readability formula is a research-proven tool, the most important predictors of text complexity are considered for ATOS and it is confirmed as valid and reliable text complexity measure.

Read-X is a web-search application to evaluate and locate potential reading material over the internet. This application searches text or keyword provided by a user on the web, it extracts text from web pages, free of HTML code and analyzes its readability using popular readability formulas. It classifies results according to thematic content and presents the thematic classification results and extracted text in an editable form (Benjamin, 2012).

Computational cohesion and coherence metrics for written and spoken texts are computed using a system named *Coh-Metrix*. Coh-metrix is used to measure the difficulty of written text for the target audience, here cohesion means characteristics of text that plays role in helping the reader to mentally connect ideas in the text. Coh-Metrix uses components of computational linguistic such as part-of-speech classifiers, syntactic parsers, lexicons and latent semantic analysis (Graesser et al., 2004).

Table 2 presents the factors, which we have termed as attributes over which the readability formulas we come across our study, uses to predict readability score or grade level. We have categorized attributes to 15 types, based on different terminologies used in the formula. As a result we found that 17 out of 21 formulas consider sentence length, 14 out of 21 formulas consider count of words and 7 out of 21 formulas consider count of syllables to result readability score or grade level. We found that attributes such as unfamiliar word, character count, familiar words, word frequency, easy words, and many more compared to sentence length, words and syllables are not considered as importantfactors to decide readability.

3. Readability of web contents & prediction

Web content is derived from various sources and there are various approaches to check its understand-ability, a common technique which can support author tools across multiple languages is presented by Nietzio et al. (2014). Realistic results for testing understand-ability could be available by asking readers for feedback, but this approach is impractical for web content. A

Table 2
Attributes of readability formulas.

Attributes	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]	[19]	[20]	[21]
Sentence	✓	✓	✓	✓	×	✓	✓	×	✓	✓	✓	×	×	✓	✓	✓	✓	✓	✓	✓	✓
Syllables	✓	×	✓	×	×	×	✓	×	✓	×	×	×	×	×	×	×	×	✓	✓	✓	×
Words	✓	✓	✓	✓	×	✓	×	×	✓	×	✓	×	✓	✓	✓	×	✓	✓	✓	✓	✓
Unfamiliar Words	×	✓	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×
Number of polysyllables word count	×	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Complex words	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	✓	✓	×	×	×	×
Character count	×	×	×	×	×	✓	×	×	×	×	✓	✓	×	✓	×	×	×	×	×	×	×
Number of periods	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×
Familiar words	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×	×
Number of sentences, number of syllables from three 100 word paragraph	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Single syllable words	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×
Average number of sentences and letters per100 words	×	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	×
Word frequency	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	✓
Easy words	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×
Paragraph	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	✓	×	×	×	×	×	×

Note: Tools [1] ⇒ FRE, [2] ⇒ Dale-Chall readability, [3] ⇒ FKGL [4] ⇒ Gunning Fog Index, [5] ⇒ SMOG, [6] ⇒ ARI, [7] ⇒ FRY, [8] ⇒ Forcast, [9] ⇒ PSK, [10] ⇒ SPACHE, [11] ⇒ CLI, [12] ⇒ BRI, [13] ⇒ LI×, [14] ⇒ Raygor Estimate Graph, [15] ⇒ Djoko Formula, [16] ⇒ New Fog Count, [17] ⇒ Pisarek's Index, [18] ⇒ Fernandez Huerta Index, [19] ⇒ Kandal and Moles Index, [20] ⇒ Al-Heeti Grade Level, [21] ⇒ Lexile.

quantitative approach to assess readability is readability indices, which quantify difficulty of text based on length and count of words, sentences and syllables. The main issue is a lot of readability formulas have been developed and indices have been created for variety of contexts but non of them is fit to be applied on contents of web. Readability indices have a lot of limitations i.e. they were developed for standard text and while writing E2R (Easy-to-Read) text, sentences become longer with more words and are classified as difficult by readability indices. Readability indices are limited to be used during the development of web content and only for the purpose of testing. To handle limitations, structure of sentences need to be considered and by applying style and grammar checkers, this can be achieved.

Readability prediction which has enormous methods to predict readability of standard text, becomes challenging to predict readability of web content as it is highly varied because of its non-traditional nature. Non-traditional nature is composed of blog, comments, search engine results, online advertising and this may also contain images, audios, videos to other rich layout elements. Labeling existing web pages with metadata that estimates readability is a solution to the problem to find a better way to search content if exists. Variety of new and surprising applications have been evolved because of labeling metadata with web pages containing readability estimation along with benefit of utility for basic web search (Collins-Thompson, 2014). Web search engines which are one of the primary ways to access information, during its design the developers does not pay much attention over readability.

Age Rank algorithm proposed by Gyllstrom and Moens (2010), aims to provide binary labeling for web documents, appropriates webpage according to child and adults, is inferred using walk algorithm. Walk algorithm is inspired by the algorithm presented by Google to estimate webpage importance named PageRank algorithm. To label pages, AgeRank approach uses features such as color of page, size of font and other additional sources which are virtue of hypertext representation also machine learning algorithms when combined with webgraph, vocabulary and non-vocabulary features provides a good basis for estimating readability (Collins-Thompson, 2014).

a **Readability methods developed using statistical language modeling:** Recent studies focuses on improving readability of web documents, resulted that noise is attributed because of availability of captions, errors in punctuation and sidebar menus. When traditional formulas were used to analyze web documents, they performed very poorly and advances in statistical language when incorporated new possibilities arisen (Collins-Thompson and Callan, 2005). Support Vector Machines(SVM) and Statistical Language Models(SLM) which were results of development in computer science and statistical models resulted for new kind of study. SLM technique is based on probability of word or words that are present in a language model for some grade level for particular stage whereas SVM helps us to identify features of grammar and patterns that are common for third grade texts, both of these develop a grade level text model and determine the likelihood of the text generated, to which model does it belongs.

In 2004–05, Collins-Thompson and Callan (Collins-Thompson and Callan, 2005) refined the technique to analyze text difficulty of web texts by using large corpus and classifying text into 12 grade-level and it was advisable to use Fry Short Passage readability formula for fourth grade and above. Towards developing high-quality formulas for the web and traditional text, researchers have found that grammatical feature sets have mixed result for web texts and adding SLM to grammatical feature moderately improves performance (Callan, 2007) and when grammatical features expanded using context free grammar parsers, the features alone performed well to predict web texts (Heilman et al., 2008). By using machine methods to determine the reading level category of a user along with analyzing the difficulty of web text, SVM method worked well in a typical search engine query (Liu et al., 2004). As online reading is becoming a universal tool for a wide variety of learners, the Online-Boost algorithm can be used to improve reading comprehension where Online-Boost is the algorithm which can package deal with the readability updating and reading comprehension evaluation (La et al., 2015), experiments conducted revealed that method proposed using this algorithm is helpful in improving learner's comprehension.

b **Correlation in advancement in cognitive theory and readability methods:** During 1970–80 when theories were emerging explaining storage and information retrieval in humans, researchers (Benjamin, 2012), who were studying text processing came up with the finding that in a text, the factors considered in the classical/traditional models are not that much contributing factors in readability as much coherence and relationship between elements of a text do contribute to readability. Because of this, researchers started analysis of difficulty of text in which theories arisen because of advancement in cognitive science were also considered as an important factor and as a result, several methods and variables were developed. Such as:

Proposition and inference model (Kintsch and Van Dijk, 1978), prototype theory (Rosch et al., 2004), latent semantic analysis (Landauer et al., 1998), semantic networks (Foltz et al., 1998) are methods which used parameters like cognitive aspects of the reader, cohesion and organization which are called as high-level parameters are introduced in text leveling methods. Computational cohesion and coherence metrics for texts written and spoken are computed using a system named Coh-Metrix. Coh-Metrix is used to measure the difficulty of written text for a target audience, here cohesion means characteristics of text that plays important role to interpret the idea in text by helping the readers to mentally connect ideas in the text. Coh-Metrix uses components of computational linguistic such as part-of-speech classifiers, syntactic parsers, lexicons and latent semantic analysis and much more.

The Delite software predicts the difficulty of text based on parameters such as morphological, lexical, syntactic, semantic and discourse. To analyze German text, a dedicated syntactic-semantic parser is used (vor der Brück et al., 2008). To normalize parameter values it incorporates machine learning algorithms which improve its performance. Predictions by Delite software are more correlated to user predictions as compared to traditional formulas and it acts as a bridge between methods motivated by cognition and statistical language modeling methods (Benjamin, 2012). The representational aspect of the textual web content i.e. lexical quality of a website can be used to measure textual web accessibility (Rello, 2012). The quality degree of words in a text such as spelling errors, typos, etc, is broadly referred as lexical quality and is related to the degree of readability of a website (Cooper et al., 2010).

Some active research is being carried out by various researchers to predict readability of web pages, which aims to improve the earlier readability formulas used to estimate readability of webpages. Article (Si and Callan, 2001) presented a statistical model to predict readability of webpages, the presented model combines readability features of text with the statistical model. The model considers content information along with linguistic features of the text and concluded that language model is a factor of importance compared to sentence length to determine the readability of webpages.

A method proposed (Yamasaki and Tokiwa, 2014) to assess readability for web documents using text features and HTML structures. Text features are extracted from text strings which form web document, features involve statistics and syntactic information of characters. HTML features include information embedded in a web browser such as headings, paragraphs, fonts, character sizes and line spacing. Vectors are used to represent the documents which are classified using machine learning, the classified web documents are used as learning data.

Researchers (Palotti et al., 2016) proposed to rank health web pages with relevance and to measure understand-ability a number readability measures were exploited and to improve search engine results readability was used. To measure syntactic and lexical features, surface measures i.e. statistics of the document such as the number of characters, syllables, words and sentences were considered. Measures related to general vocabulary i.e. common lexical features used to measure text difficulty. The proportion of numbers, stop-words and common words in the document were treated as lexical features. Medical vocabulary related measures which are specially adapted in the scientific domain were used as lexical and morphological features. It was found (Palotti et al., 2016) that combining retrieval features with readability features improves search engine results.

GUI Evaluator (Alemerien and Magel, 2014), a tool to measure the complexity of graphical user interface based on structural measures of information complexity such as alignment, grouping, size, density and balance. The graphical features of the web can be quantified using this approach, and the effect of graphics can also be taken into account while predicting readability. Following are the attributes of GUI Evaluator:

i. Alignment: There are two levels of measurement namely Group level called Local Alignment and the Screen Level called Global Alignment, are used to measure the vertical and horizontal alignment of objects. The Total Alignment complexity (TAC) is given by Eq. (1), in which weight1 is the ratio of a number of grouped objects to the total number of objects on screen, and weight2 represents the ratio of the number of ungrouped objects to a total number of objects on screen.

$$\begin{aligned}
&TAC = AC * weight1 + SA * weight2\\
&Where,\\
&SA = \frac{\sum_{l=1}^{N}(Vp + Hp)}{2N},\\
&AC = \sum_{i=1}^{m}(GAi * Weight),\\
&GAi = \frac{\sum_{l=1}^{K}(Vp + Hp)}{2K}
\end{aligned} \tag{1}$$

ii. Balance: The number and size of objects are used for Balance metric and Total Balance Complexity (TBC) is calculated as shown in Eq. (2) for each quarter of screen where BQni and BQnj variables represent the number of objects in ith and jth quarters. The overall value of BQn variable ranges within [0,1] it (where 0 means unbalanced and 1 means fully balanced in terms of number of objects). Similarly, BQsi and BQsj represents the sum of sizes of objects in ith and jth quarters respectively, range of ratio of BQsi and BQsj varies between [0,1] and overall values of BQs is [0,1].

$$\begin{aligned}
&TBC = 1 - (0.5 * BQn + 0.5 * BQs)\\
&Where,\\
&BQn = \frac{\sum_{k=1}^{6}\frac{BQni}{BQnj}}{6}\\
&BQs = \frac{\sum_{k=1}^{6}\frac{BQsi}{BQsj}}{6}
\end{aligned} \tag{2}$$

iii. Density: Density Complexity (DC) is calculated considering W1 (ratio of area of groups to screen) and W2 (ratio of ungrouped area to the screen area). It meaures the screen occupation by objects calculating Local Density (LDj) for jth group and Global Density (GD).

$$\begin{aligned}
&DC = \left(\frac{\sum_{j=1}^{n}LDj}{n}\right) * W1 + GD * W2\\
&Where,\\
&GD = \frac{\sum_{k=1}^{\text{ungrouped objects}}\text{Size of object k}}{\text{Area of ungrouped}},\\
&LDj = \frac{\sum_{i=1}^{\text{grouped objects}}\text{Size of Object i in a group j}}{\text{Area of group j}}
\end{aligned} \tag{3}$$

iv. Size: Object size complexity is measured at two levels using following size metrics.

$$SCk = \frac{\sum_{j=1}^{N} Sj}{N}$$
$$SC = \frac{\sum_{k=1} WiSCk * Weight(k)}{Wi} \quad (4)$$

N is number of objects in type kth and Sj is number of different sizes i.e value of Sj is 1 if object size is not counted and 0 if object size is counted.

v. Grouping: The measurement of number of objects having clear boundary by line,color,background,size or space is measured using Grouping Metric.

$$GT = UG + GC$$
$$GC = \frac{G}{M} * Weight \quad (5)$$
$$UG = 1 - \frac{\sum_{i=1} NGW}{N}$$

where GW represents object that is grouped and Eq. (12) calculates ratio of number of different object types (G) to total number of objects (M) in all groups,the ratio of grouped objects to total number of objects on screen is represented by Weight.

vi. Overall Screen Layout Complexity (LC) is given as below:

$$\begin{aligned} &LC = ((A + B + C + D + E)/5) * 100 \\ &Where, \\ &A = TAC * w1, \\ &B = TBC * w2, \\ &C = DC * w3, \\ &D = SC * w4, \\ &E = GT * w5 \end{aligned} \quad (6)$$

where w1,w2,w3,w4,w5 represents respective weights of complexities having values 0.84,0.76,0.80,0.72 and 0.88 respectively. An application developed to collect data, initially, participants were explained how to rate the design factors than they were asked to provide background information and finally asked to rate the user interface design and above discussed five design factors to measure the complexity.

Section 3 of this paper discusses developments and work done in the field to predict readability of web contents. We assessed 19 different articles as presented in (Table 3) to know which readability tool/model is mostly used to predict readability of various readable materials. Flesch Reading Ease and Flesch-Kincaid Grade level models have been mostly used(almost 42%, which is very high compared to other formulas), to predict the readability of texts or written materials. Among the articles studied (8 out 0f 19 almost 42%) articles aimed to study the readability of the web and the web related contents and we found that the tools used to predict readability of traditional text are used to predict the readability of text on the web.

Because of ease of using these tools, the formulas are adapted to predict readability but we cannot rely on the readability scores resulted as readability of websites and web pages depends on many factors which these traditional formulas don't consider while computing readability score.

4. Readability prediction of contents for people with disabilities

In our survey enormous matter discussing understandability of text for people with different disabilities was not found. A lot of research has been done to enhance readability for normal people. When classic readability features were examined for reading items to identify reading difficulties because of grammatical and cognitive features, the negative effect was found on reading performance of people with disabilities (Abedi et al., 2012). Surface textual/visual features such as long words, font, word length and spacing are the important factors which had the highest discriminative power between people with and without disabilities (Collins-Thompson, 2014). Similar findings were made by Rello et al. (2012) for readers with dyslexia. Automated readability assessment tools were developed and evaluated for readers with intellectual disabilities and explored the use cognitively-motivated features, for example, the number of entities mentioned per sentence (Feng et al., 2009), the readability of disabled learners can be improved by using techniques to simplify and summarize the text. Set of protocols which assist developers in identifying characteristics and problems faced by visually impaired people were presented in Ferreira et al. (2012), similar protocols need to developed to analyze and improve readability of people with special needs by involving them during design. As access to graphics by visually impaired people is recognized as an area of difficulty, a study focused on understanding the challenges faced by visually impaired students while accessing graphics on websites was conducted (Butler et al., 2016). The findings of this article will be helpful for the developers while developing web contents.

5. Conclusion and discussion

In this paper, a lot of readability indices and tool based research are reviewed, which are used and published in the area of understand-ability of web contents. These readability formulas and tools have been developed to measure the complexity of the traditional text. But, a lot of factors which need to be analyzed while measuring readability properties of the web such as the relationship of one meta-data with other for the same page, analysis of differences in reading level distributions across different domains and web pages. These models do not consider knowledge or user's context while assigning a label for readability.

However, the readers have to face a lot of new contents such as blogs, wiki, and other web contents, web interactions especially for educational settings play an important role. So, readability methods and formulas which have been evolved still do not focus on the effects of these contents on readability.

The recently developed methods which are based on cognitive load and statistics of language more or less they are dependent on traditional formulas, which consider only traditional contents of readability.

In 2014, K. Collins-Thompson carried out the computational assessment of text readability which indicates that future research on readability needs to be user-centric, data-driven and knowledge based (Collins-Thompson, 2014).

The rise of new readability formulas aims to overcome past weaknesses. In addition, by the use of natural language processing, statistical methods, and other computerized approaches, we will be able to assess readability accurately. But still, the readability formulas only focus on English texts. These formulas will not efficiently work on other languages which use scripts other than used by English.

Now, web pages are multi-lingual so the readability formulas should work accurately to fit the web contents along with normal text too, this needs to be taken into account. A lot of research is needed to make the readability measures adaptable for the web contents and text developed, which focuses on the accessibility-cum-readability-cum-understandability for people with different kinds of disabilities. To make the computation of readability score for web pages better, some of the web specific measures which can

Table 3
Survey of articles assessing readability.

S.No	Author Name	Test Data	FRE	DC	FKGL	GFOG	SMOG	CLI	ARI	Other Models
1	Lei Lei, Sheng Yan (2016)	Mean of Readability scores of articles published in four Journals of Information Science (from 2003 to 2012) (Lei and Yan, 2016)	−29.26 (difficult abstract) 78.25 (easy abstract)	×	×	×	26.72 (difficult abstract) 9.71 (easy abstract)	×	×	×
2	Shodi Fallah, Sepideh Rahimpour	Readability scores of translated scientific text by 3 groups of translators into persian text (Fallah and Rahimpour, 2016)	SaT-43 GT-50.3 ST-46.3	×	SaT-12.3 GT-9.9 ST-10.4	SaT-16.2 GT-14.6 ST-14.5	SaT-12.2 GT-10.6 ST-11	×	×	×
3	Ben Seipel et al.	Average grade level of 40 original Multiple Choice Online Cloze Comprehension Assesment (MOCCA) test items	×	×	Mean-4.43	×	×	×	×	×
4	Pratik Shukla et al.	Mean of readability scores of 40 UAE (uterine artery embolization) related Internet based patient education materials created by US hospitals (Shukla et al., 2013)	43.98	×	10.96	14.55	13.63	×	×	×
5	Rend Alkhalili et al.	Mean of readability scores of Internet based patient education materials related to mammography for breast cancer screening (AlKhalili et al., 2015)	38.42 to 59.66	×	8.7 to 12.72	12.14 to 16.5	11.66 to 15	×	×	×
6	Tian Kang et al.	Readability scores of health texts available on ClinicalTrial.gov for clinical trial seekers. (Kang et al., 2015)	×	11.22	×	×	14.09	×	×	×
7	Jane P. Wayland Cynthia M. Daily	Range of readability scores of research papers and text books related to marketing. (Wayland and Daily, 2015)	Research Papers: [23.7–32.6] Textbooks: [28.3–39.7]	×	×	×	×	×	×	×
8	Samuel Severance K. Brit onnelCohen	Mean of readability scoresof abstracts of medical research journals from 1960 to 2010. (Severance and Cohen, 2015)	×	×	×	×	×	1960s-16.08 1970s-16.32 1980s-16.38 1990s-16.43 2000s-16.86	×	×
9	Sara Dolnicar, Alexander Chapple	Average of readability scores of 493 articles of tourism journals (Dolnicar and Chapple, 2015)	18.0	×	×	×	×	×	×	×
10	Taha Yasseri et al.	Mean of readability indices of selected 10 articles from different topical categories for Simple text and Main text (Yasseri et al., 2012)	×	×	×	Simpletext [9.04] Maintext [13.37]	×	×	×	×
11	Osman Cardak et al.	Average readability score of 7 text samples from 7th grade science textbook published in Turkish language (Cardak et al., 2015)	×	×	23.72	26.96	×	×	×	Sonmez Formula [0.0000966]very easy Cloze Test Method [45.201]
12	Muhammad Kunta Biddinka et al.	Average Readability of 19 Biomass websites of Indonesia (Biddinika et al., 2016)	×	×	20.14	24.67	16.384	×	×	×
13	Abdel karim Al Tamimi et al.	Average grade level for various texts of elementary classes from Jordanian curricullum (Al Tamimi et al., 2014)	×	×	×	×	×	×	7.4382	LIX-22.61 Al-Heeti gradelevel-5.35
14	Renu Gupta	Readability of English text-books used in primary classin English medium schools in India(Gupta, 2014)	×	×	×	×	×	×	×	Coh-metrix to check narrativity NCERT [78–94] Gulmohar [67–98] Oxford [29–89] Images [61–95]

Table 3 (*continued*)

S.No	Author Name	Test Data	FRE	DC	FKGL	GFOG	SMOG	CLI	ARI	Other Models
15	James R.A. Daven-port, Robert Deline	Readability of tweets SMS and Chats (Davenport and DeLine, 2014)	Tweets-[50.80] SMS-[88.2] Chats-[54.0]	×	×	×	×	×	×	×
16	Ali Akbar Jabbari Nazanin Saghari	Difficulty level of 50 medical booklets translated to Persian from English language	×	×	×	Persian [18.056] English [13.522]	Persian [8.395] English [7.031]	×	×	×
17	Xiaoyong Liu et al.	10–12 + grad level readability scores of 20 websites resulted from user level queries to search search engines reading	×	×	11.53	0	0	×	×	×
18	Alexandra H. Humphreys et al.	Mean of difficulty level of selected 22 articles in the journal of Music Education and journal of Historical Research in Music Education	38.65	11.55	13.75	16.3	14.6	×	×	×
19	Cherin C. Pace et al.	Mean of readability scores of 10 published patient related outcome (PRO) questionnaires related to oral health quality of life	77.7	×	×	7.7		×	×	Forcast = 9.1

be incorporated into the computation of readability score, are as listed below:

1. The structural components present in the web pages such as hyperlinks, alt-text used for images shall be incorporated into the computation of readability of web pages.
2. A variable weight based approach can be adopted for different elements of webpages. For example, the text present in the Level 1 heading can be associated with higher weight than an normal text.
3. Another important aspect of web is dynamism. In webpages, it is not required that all content need to be visible at all times. Certain content will be visible only when user performs certain action (such as a "+" symbol). Hence, the readability score computation methods for web pages can be built to incorporate such dynamism into consideration, instead of including all text at once.
4. As web pages generally contains shorter text elements, there is a need for web-specific versions for existing readability scoring formulae.

The primary focus of this paper was to provide a detailed insight into readability measurement. The specific requirements of web content readability were also presented. The correlation between accessibility and readability was also highlighted. Some specific measures that can be adopted while computing web page content readability were also listed. To conclude, it becomes mandatory for the content providers to focus more on the quality of text along with the style with which it is presented, to make it universally accessible.

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